



Quiz # 6 - Solution

Name: _____

Roll # _____

Content	Clo's	Activity

Write SQL (Joins) Queries

1. From the Customer and Salesman tables write a SQL query to find the salesperson(s) and the customer(s) he represents. Return Customer Name, city, Salesman, commission.

```
SELECT a.cust_name AS "Customer Name",  
       a.city,  
       b.name AS "Salesman",  
       b.commission  
FROM customer a  
INNER JOIN salesman b  
ON a.salesman_id = b.salesman_id;
```

2. From the Customer and Salesman tables write a SQL query to locate those salespeople who do not live in the same city where their customers live and have received a commission of more than 12% from the company. Return Customer Name, customer city, Salesman, salesman city, commission.

```
SELECT a.cust_name AS "Customer Name",  
       a.city,  
       b.name AS "Salesman",  
       b.city, b.commission  
FROM customer a  
INNER JOIN salesman b  
ON a.salesman_id = b.salesman_id  
WHERE b.commission > 0.12  
AND a.city <> b.city;
```



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3. From the Customer and Orders tables write a SQL query to find those orders where the order amount exists between 500 and 2000. Return ord_no, purch_amt, cust_name, city.

```
SELECT a.ord_no, a.purch_amt, b.cust_name, b.city
FROM orders a, customer b
WHERE a.customer_id = b.customer_id
AND a.purch_amt BETWEEN 500 AND 2000;
```

4. From the Customer, Salesman and Orders tables write a SQL query to find the details of an order. Return ord_no, ord_date, purch_amt, Customer Name, grade, Salesman, commission.

```
SELECT a.ord_no, a.ord_date, a.purch_amt,
       b.cust_name AS "Customer Name", b.grade,
       c.name AS "Salesman", c.commission
FROM orders a
INNER JOIN customer b
ON a.customer_id = b.customer_id
INNER JOIN salesman c
ON a.salesman_id = c.salesman_id;
```

5. Write a SQL statement to join the tables salesman, customer and orders so that the same column of each table appears once and only the relational rows are returned.

```
SELECT *
FROM orders
NATURAL JOIN customer
NATURAL JOIN salesman
```

6. From the Customer and Salesman tables write a SQL query to display the customer name, customer city, grade, salesman, salesman city. The results should be sorted by ascending customer_id.

```
SELECT a.cust_name, a.city, a.grade,
       b.name AS "Salesman", b.city
FROM customer a
LEFT JOIN salesman b
ON a.salesman_id = b.salesman_id
ORDER BY a.customer_id;
```



7. From the Customer and Salesman tables write a SQL statement to generate a list in ascending order of salespersons who work either for one or more customers or have not yet joined any of the customers.

```
SELECT a.cust_name, a.city, a.grade,  
       b.name AS "Salesman", b.city  
FROM customer a  
RIGHT OUTER JOIN salesman b  
ON b.salesman_id = a.salesman_id  
ORDER BY b.salesman_id;
```

8. From the Customer and Order tables write a SQL statement to make a report with customer name, city, order number, order date, and order amount in ascending order according to the order date to determine whether any of the existing customers have placed an order or not.

```
SELECT a.cust_name, a.city, b.ord_no,  
       b.ord_date, b.purch_amt AS "Order Amount"  
FROM customer a  
LEFT OUTER JOIN orders b  
ON a.customer_id = b.customer_id  
ORDER BY b.ord_date;
```

9. Write a SQL query to combine each row of the salesman table with each row of the customer table.

```
SELECT * FROM salesman a  
  
CROSS JOIN customer b;
```

10. Write a SQL statement to create a Cartesian product between salesperson and customer, i.e. each salesperson will appear for all customers and vice versa for that salesperson who belongs to that city.

```
SELECT *  
FROM salesman a  
CROSS JOIN customer b  
WHERE a.city IS NOT NULL  
AND b.grade IS NOT NULL  
AND a.city <> b.city;
```



Samples Tables

Sample table: customer

3002	Nick Rimando	New York	100	5001
3007	Brad Davis	New York	200	5001
3005	Graham Zusi	California	200	5002
3008	Julian Green	London	300	5002
3004	Fabian Johnson	Paris	300	5006
3009	Geoff Cameron	Berlin	100	5003

Sample table: salesman

salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13

Sample table: orders

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.5	2012-10-05	3005	5002
70009	270.65	2012-09-10	3001	5005
70002	65.26	2012-10-05	3002	5001
70004	110.5	2012-08-17	3009	5003
70007	948.5	2012-09-10	3005	5002